

Wave 5.2R Stage 0 Forward Evidence Freeze

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Summary

Stage 0 is complete and passes its exit gate.

The repository now has one immutable, forward-only baseline contract for `polished_dataset` with setpoint inputs. The contract covers the same `97` eligible held-out `Fw` operating conditions for:

- `PF_A_LOCAL_QUADRATIC` ;
- `accepted_periodic_mlp_harmonic_Fw` ;
- `accepted_periodic_gru_sequence_Fw` .

The split signature is `c1aa8718fb9bf88cc2021c121dc4f3b4010fc1d2e45ac90af5f4376aa64f8e16` . All twelve numerical reproduction checks pass within their declared tolerances. No training was executed and no test condition entered the Polynomial-Fourier coefficient fit. The forward GRU remains the lowest raw-error reference. The harmonic MLP has the best current composite curve-payload diagnostic score because it preserves the selected harmonic amplitudes better on average. `PF-A` remains a useful, inspectable analytical anchor, but it is not the best raw predictor.

Scope And Evidence Contract

Field	Frozen value
Dataset	<code>polished_dataset</code>
Input mode	<code>setpoints</code>
Surface	<code>Fw</code> only
Eligible train conditions	<code>675</code>
Eligible validation conditions	<code>194</code>
Eligible test conditions	<code>97</code>
Quarantined anomalies	<code>3</code> training conditions
Angular unit	degree
Evaluation resolution	full curve
Training in this stage	none

The condition manifest preserves the source path, byte size, and SHA-256 of every eligible forward curve. The candidate provenance inventory also hashes the neural checkpoints, model inventories, split evidence, replay outputs, and Polynomial-Fourier coefficient artifacts used by the freeze.

Reproduction Result

PF-A reproduces the canonical Phase 1 forward test metrics exactly to the stored floating-point precision. Its four checked metric differences are 0.0 , against an absolute tolerance of 1e-12 degrees.

The two neural references reproduce the earlier CVP 1.2 evidence within:

- 1e-6 degrees for mean curve MAE;
- 0.01 percentage points for percentage and amplitude metrics;
- 0.01 degrees for mean phase error.

The largest observed neural difference is 0.001085 degrees in the GRU mean harmonic phase error. This is below tolerance and is consistent with the current diagnostic implementation rather than a model or split change.

Baseline Comparison

Candidate	Mean MAE [deg]	P95 MAE [deg]	Mean RMSE [deg]
PF_A_LOCAL_QUADRATIC	0.001807	0.003942	0.002120
Harmonic MLP Fw	0.001694	0.004098	0.002008
Sequence GRU Fw	0.001618	0.003817	0.001931

Candidate	Centered MAE [deg]	Absolute mean error [deg]
PF_A_LOCAL_QUADRATIC	0.001385	0.000965
Harmonic MLP Fw	0.001390	0.000825
Sequence GRU Fw	0.001382	0.000690

The GRU is the strongest entry reference for raw curve error, upper-tail error, centered shape, and mean-offset error. PF-A narrowly beats the harmonic MLP on centered MAE, but trails both neural references on raw and offset metrics.

The Phase 1 and CVP 1.2 metric contracts remain explicit. In particular, PF-A reports retained-coefficient amplitude error in degrees and phase error in radians, whereas CVP 1.2 reports relative amplitude error in percent and phase error in degrees. These fields must not be ranked directly across the analytical and neural contracts.

Curve-Payload Diagnostics

The derivative metric in the following table is expressed in degrees per degree.

Candidate	Mean MPE [%]	P2P error [%]	Derivative RMSE
Harmonic MLP Fw	3.438600	11.801784	0.014963
Sequence GRU Fw	3.278427	8.443678	0.014968

Candidate	Harmonic amplitude error [%]	Harmonic phase error [deg]	Diagnostic score
Harmonic MLP Fw	15.380051	13.591649	5.805821
Sequence GRU Fw	19.236860	12.165789	5.960079

The MLP wins the diagnostic score because the score penalizes mean harmonic amplitude error. The GRU is still better on raw percentage error, peak-to-peak behavior, and mean phase error. This split result is why Stage 0 freezes both accepted neural references instead of collapsing them into one winner.

Per-order exports cover harmonics 1, 3, 39, 40, 78, 81, 156, 162, and 240. Harmonic 0 is represented by the separate mean and offset channels rather than a phase comparison.

Operating-Cell Evidence

The normalized operating-cell table contains 291 rows:

- 97 for PF-A ;
- 97 for the harmonic MLP;
- 97 for the sequence GRU.

Each row retains speed, torque, oil temperature, raw MAE, RMSE, centered MAE, and absolute curve-mean error. This table is the Stage 0 input for later worst-cell, regime, robustness, and residual analyses.

Leakage And Provenance Audit

The no-test-fit gate passes for three independent reasons:

- the Polynomial-Fourier replay declares `training_executed: false` ;
- its coefficient artifact declares `fit_scope: eligible training conditions only` ;
- both neural candidates are pre-existing exported Wave 1 models evaluated by an inference-only comparison runner. The fresh Polynomial-Fourier replay and neural replay write into isolated Stage 0 directories and do not overwrite the earlier Phase 0, Phase 1, or Phase 2 evidence.

Exit Gate

Check	Result
Split identity	Pass
Eligible condition counts	Pass
Exactly 97 test curves	Pass
Exact three-candidate roster	Pass
Forward-only evidence	Pass
Finite metrics	Pass
Reproduction tolerances	Pass
No test data in fit state	Pass

Decision: accept the Stage 0 evidence freeze and authorize Stage 1, Extended Scientific Technique Discovery. This decision authorizes research and documentation only. Any later training stage still requires its campaign planning package and campaign approval gate.

Machine-Readable Artifacts

The canonical freeze root is

```
output/analysis/wave_5_2r/stage0_forward_evidence_freeze/frozen_contract/
```

It contains

- `stage0_forward_evidence_freeze.yaml` ;
- `forward_condition_manifest.csv` ;
- `baseline_metrics.csv` ;
- `operating_cell_metrics.csv` ;
- `harmonic_band_metrics.csv` ;
- `provenance.csv` ;
- `reproduction_comparison.csv` .

The full neural curve payloads remain under

`output/validation_checks/wave52r_stage0_forward_curve_payload_diagnostics/` . The isolated PF-A replay remains under `output/analysis/wave_5_2r/stage0_forward_evidence_freeze/pf_a_reproduction/` .

Reproduction Commands

Build and validate the frozen contract with:

- `conda run --no-capture-output -n pinns_env python -B scripts/analysis/wave_5_2r/stage0_forward_evidence_freeze/build_stage0_forward_evidence_freeze.py` ;
- `conda run --no-capture-output -n pinns_env python -B scripts/analysis/wave_5_2r/stage0_forward_evidence_freeze/validate_stage0_forward_evidence_freeze.py` .

The replay configs and exact evidence paths are stored under

`config/analysis/wave_5_2r/stage0_forward_evidence_freeze/` .

Conclusion

Stage 0 removes ambiguity from all later wave 5.2R comparisons. The forward GRU is the raw-error reference, the harmonic MLP is the current harmonic-diagnostic reference, and PF-A is the analytical anchor. Later stages must beat matched controls on this exact test surface and must preserve the contract-specific units recorded here.